

Insulin secretion as a determinant of pancreatic cancer risk.

New epidemiology confirms that glucose intolerance is a risk factor for pancreatic cancer, and that this association cannot be accounted for by an adverse impact of early pancreatic cancer on beta cell function. Previous reports indicate that risk for pancreatic cancer is increased in adult-onset diabetics. Since streptozotocin diabetes inhibits carcinogen-mediated induction of pancreatic cancer in hamsters, the most reasonable interpretation of these findings is that insulin (or some other beta cell product) acts as a promoter for pancreatic carcinogenesis. This view is consistent with a report that human pancreatic adenocarcinomas express insulin receptors that can stimulate mitosis; an additional possibility is that high insulin levels indirectly promote pancreatic carcinogenesis by boosting effective IGF-I activity via hepatic actions. In international ecologic epidemiology, pancreatic cancer rates correlate tightly with dietary intake of animal products; this may reflect the fact that vegan diets are associated with low diurnal insulin secretion. There is also suggestive evidence that macrobiotic vegan diets, which are low in glycemic index, may increase mean survival time in pancreatic cancer. However, other types of diets associated with decreased postprandial insulin response, such as high-protein diets or 'Mediterranean' diets high in oleic acid, may also have the potential for pancreatic cancer prevention. The huge increases of age-adjusted pancreatic cancer mortality in Japan and among African-Americans during the last century imply that pancreatic cancer is substantially preventable; a low-insulin-response diet coupled with exercise training, weight control, and smoking avoidance, commendable for a great many other reasons, may slash pancreatic cancer mortality dramatically. Copyright 2001 Harcourt Publishers Ltd.